SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Data Science

ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA17	**Course Title:**	Artificial Intelligence
CO Assessed:	CO1 – Apply (BL3)	**Assessment:**	Assessment Tool 1 – Analytical Problem Solving
Weightage:	35% of CIA	**Duration / Marks:**	60 mins
CO1 Statement:	Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)		
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Section / Batch:	2025-2029	**Date:**	3.10.2026

Code of Conduct

I KEERTHANA R (192524489) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: KEERTHANA R

Instructions

- Read each question carefully before answering.
- Answer all four questions. Show every step: state your assumptions, the method you choose and why, your calculations, and finally interpret the result for the agent or AI system in the problem.
- Questions are drawn from Foundations and History of AI, Intelligent Agents (Agents and Environments) and the Concept of Rationality.
- Prepare digital Write up units and clear final answers (marks, minutes, percentage points, expected utility); present tables and working neatly.
- Marks will be awarded using the Analytical Problem Solving rubric: problem understanding, method selection, solution process, accuracy of calculation and interpretation of results.
- Upload in Github and GCR (Question paper with regNo, Name, Answers and rubrics and evalution page at last).
- Duration: 1 hour (15 minutes per question).

Section / Topic	Focus	Question Nos.
1. Foundations and History of AI	Definitions of AI (four approaches), Turing Test, eras and milestones in the history of AI	**Q3**
2. Intelligent Agents: Agents and Environments	Agent function and program, PEAS description, task environment properties, agent architectures	**Q1, Q2**
3. Concept of Rationality	Performance measure, percept sequence, expected utility, rational versus omniscient agent	**Q1, Q4**

Question 1: Two-Room Vacuum World

Topic: Agents and environments, concept of rationality | 15 minutes

A vacuum-cleaner agent lives in a world with two squares, A and B. At each time step it perceives its own location and whether that square is Dirty or Clean. Its actions are Suck, Right, Left and NoOp. The agent starts in square A, and both squares are initially Dirty. Once a square is cleaned, dirt never reappears.

Performance measure over a lifetime of T = 6 time steps: +1 for every clean square at the end of each time step, and a penalty of p for every Left or Right move (take p = 1 unless told otherwise).

- **Agent X (simple reflex):** if the current square is Dirty, Suck; otherwise move to the other square.
- **Agent Y (reflex with internal state):** if the current square is Dirty, Suck; otherwise, if it remembers that both squares have been cleaned, NoOp; otherwise move to the other square.

(a) Write the PEAS description (Performance measure, Environment, Actuators, Sensors) of this agent from the data given. **[4 Marks]**

The vacuum world contains two squares, A and B. The agent starts in A, and both squares are initially dirty.

- **Performance Measure:**
  The agent gets +1 for every clean square at the end of each time step. A penalty of p is applied whenever the agent performs Left or Right.
- **Environment:**
  Two squares, A and B, where each square can be Dirty or Clean. Once cleaned, a square remains clean.
- **Actuators:**
  Suck, Left, Right and NoOp.
- **Sensors:**
  The agent can perceive its current location and whether the current square is Dirty or Clean.

Thus, the main objective is to keep both squares clean while avoiding unnecessary movement.

(b) Trace Agent X for all 6 time steps in a table (action, clean squares at end of step, move penalty, step score) and find its total score for p = 1. **[5 Marks]**

Agent X is a simple reflex agent. Its rule is:

- If the current square is Dirty → Suck.
- If the current square is Clean → Move to the other square.

The agent starts at A and both squares are dirty.

Step	Action	Clean Squares	Step Score
1	Suck	A	1
2	Right	A	0
3	Suck	A, B	2
4	Left	A, B	1
5	Right	A, B	1
6	Left	A, B	1

For p = 1:

Total Score = 1 + 0 + 2 + 1 + 1 + 1 = 6

Therefore, **Agent X's total score = 6**.

Agent X cleans both squares, but after cleaning them it continues moving because it has no memory of previous actions.

(c) Trace Agent Y in the same way and find its total score for p = 1. **[4 Marks]**

Agent Y is a reflex agent with internal state. It remembers that both squares have been cleaned. Once both are clean, it performs NoOp.

Step	Action	Clean Squares	Step Score
1	Suck	A	1
2	Right	A	0
3	Suck	A, B	2
4	NoOp	A, B	2
5	NoOp	A, B	2
6	NoOp	A, B	2

For p = 1:

Total Score = 1 + 0 + 2 + 2 + 2 + 2 = 9

Therefore, **Agent Y's total score = 9**.

Agent Y performs better because it avoids unnecessary movement after both squares are clean.

(d) Derive the total score of X and of Y as functions of p. For which values of p is Y strictly better than X? **[3 Marks]**

The total clean-square contribution over six steps is:

1 + 1 + 2 + 2 + 2 + 2 = 10

Agent X makes four movements. Therefore:

Score(X) = 10 − 4p

Agent Y makes only one movement. Therefore:

Score(Y) = 10 − p

For Y to be strictly better:

10 − p > 10 − 4p

3p > 0

Therefore:

Y is strictly better than X when p > 0.

When p = 0, both agents have the same score.

(e) Which agent is rational? Interpret your result: explain how the performance measure and the percept history decide rationality, and why X cannot match Y even though both see the same percepts at each step. **[4 Marks]**

- Agent Y is more rational when **p > 0** because it chooses actions that maximize the given performance measure.
- Both agents receive the same current percepts, but Agent Y also maintains an internal state. After both squares are cleaned, Y remembers this information and chooses NoOp instead of moving.
- Agent X cannot do this because it depends only on the current percept. Therefore, it continues moving whenever it sees a clean square.
- This shows that rationality depends on the **performance measure, percept history and information available to the agent**.

Question 2: Task Environment Analysis and Agent Selection

Topic: Agents and environments (PEAS, environment properties, agent types) | 15 minutes

Three AI agents are to be deployed in the following task environments:

- **E1:** An autonomous taxi driving through city traffic.
- **E2:** A part-picking robot that uses a camera to pick defective parts off a moving conveyor belt and place them in bins.
- **E3:** A chess-playing agent in a tournament game played with a clock.

Difficulty score convention: for each of the six properties below, the harder value scores 1 point and the easier value scores 0. A semi-dynamic environment scores 0.5 on the Static/Dynamic property.

Property	Easier value (0)	Harder value (1)
Observability	Fully observable	Partially observable
Number of agents	Single agent	Multi-agent
Determinism	Deterministic	Stochastic
Episodic / Sequential	Episodic	Sequential
Static / Dynamic	Static	Dynamic (semi-dynamic = 0.5)
Discrete / Continuous	Discrete	Continuous

(a) Classify each environment on all six properties with a one-phrase justification for each. **[9 Marks]**

E1 – Autonomous Taxi

The taxi environment is **partially observable** because the taxi cannot observe every road condition, vehicle or pedestrian.
It is **multi-agent** because other drivers and pedestrians also affect the environment.
It is **stochastic** because traffic behaviour and road conditions are uncertain.
It is **sequential** because one driving decision can affect future decisions and states.
It is **dynamic** because traffic conditions continuously change.
It is **continuous** because position, speed, steering and time can change continuously.
Therefore:
E1 = Partially observable, multi-agent, stochastic, sequential, dynamic and continuous.

E2 – Part-Picking Robot

The robot is **partially observable** because the camera may not provide complete information about every part.
It is **single-agent** because the robot itself makes the decisions.
It is **stochastic** because detection and picking can involve uncertainty.
It is **episodic** because each defective part can be treated as a separate picking task.
It is **dynamic** because the conveyor belt is continuously moving.
It is **continuous** because the position and movement of parts vary continuously.

Therefore:
E2 = Partially observable, single-agent, stochastic, episodic, dynamic and continuous.

E3 – Chess

Chess is **fully observable** because the complete board position is visible.
It is **multi-agent** because two players participate.
It is **deterministic** because a legal move produces a definite resulting board state.
It is **sequential** because every move affects future moves.
It is **semi-dynamic** because the chess position remains unchanged while thinking, but the game clock continues.
It is **discrete** because the board positions and possible moves are discrete.
Therefore:
E3 = Fully observable, multi-agent, deterministic, sequential, semi-dynamic and discrete.

(b) Compute the difficulty score of E1, E2 and E3 and rank them from hardest to easiest. **[4 Marks]**
Using the given scoring convention:
E1 – Taxi

- Partially observable = 1
- Multi-agent = 1
- Stochastic = 1
- Sequential = 1
- Dynamic = 1
- Continuous = 1

Total = 6
E2 – Robot

- Partially observable = 1
- Single-agent = 0
- Stochastic = 1
- Episodic = 0
- Dynamic = 1
- Continuous = 1

Total = 4
E3 – Chess

- Fully observable = 0

- Multi-agent = 1
- Deterministic = 0
- Sequential = 1
- Semi-dynamic = 0.5
- Discrete = 0

Total = 2.5
Therefore, from hardest to easiest:
E1 → E2 → E3
The difficulty scores are:
E1 = 6, E2 = 4, E3 = 2.5.

(c) For the hardest and the easiest environment, choose the most suitable agent architecture (simple reflex, model-based reflex, goal-based, utility-based, learning) and justify the choice using your classification. **[4 Marks]**

E1 – Autonomous Taxi
A **utility-based agent** is suitable for the taxi.
The taxi must consider several factors such as safety, travel time, traffic and uncertainty. A utility-based agent can compare possible actions and choose the one that gives the highest expected utility.
E3 – Chess
A **goal-based agent** is suitable for chess because there is a clear goal and the environment is fully observable and deterministic.
The agent can evaluate possible moves and select moves that help it achieve the goal of winning the game.

(d) Interpret: explain how the properties of an environment, not just its domain, decide how complex the agent program must be. **[3 Marks]**

- The complexity of an intelligent agent depends on the properties of its environment, not simply on the application area.
- A taxi operates in a partially observable, stochastic, multi-agent, dynamic and continuous environment, so it needs more complex decision-making.
- Chess is fully observable and deterministic, making its environment easier to model.
- Therefore, **environment properties directly influence the design and complexity of the agent program.**

Question 3: Turing Test Data Across the History of AI

Topic: Foundations and history of AI (Turing Test, acting humanly, eras of AI) | 15 minutes

In a Turing-test style study, judges held five-minute text conversations with an unknown party and then said whether it was a human or a machine. Each system below faced 50 independent interrogations in which it was the machine. The table shows how many times the judge correctly identified it as a machine.

System	Era / technique	Interrogations	Correctly identified
ELIZA	1960s, keyword pattern matching	50	47
Chatbot B	2010s, scripted rules plus statistical NLP	50	36
System C	2020s, large neural language model	50	29

Reference: Turing (1950) predicted that an average interrogator would have no more than a 70% chance of making the right identification after five minutes. A judge who is purely guessing is right 50% of the time.

(a) Compute the correct-identification rate and the fooling rate (percentage of interrogations in which the machine was taken for a human) for each system. **[6 Marks]**

The formulas are:

Correct-identification rate = Correctly identified / Total × 100

Fooling rate = (Total − Correctly identified) / Total × 100

ELIZA

Correct-identification rate:

47/50 × 100 = 94%

Fooling rate:

(50 − 47)/50 × 100 = 6%

Chatbot B

Correct-identification rate:

36/50 × 100 = 72%

Fooling rate:

(50 − 36)/50 × 100 = 28%

System C

Correct-identification rate:

29/50 × 100 = 58%

Fooling rate:

(50 − 29)/50 × 100 = 42%

Therefore:

System	Correct Identification	Fooling Rate
ELIZA	94%	6%
Chatbot B	72%	28%
System C	58%	42%

(b) Decide which systems satisfy Turing's 70% criterion. State the method you use and how far each system is from the 50% chance level. **[5 Marks]**

The stated criterion is that an average interrogator would have no more than a **70% chance of correctly identifying the machine**.

- For ELIZA:

94% > 70%

Therefore, ELIZA does not satisfy the criterion.

Distance from 50% guessing:

94 − 50 = 44 percentage points

- For Chatbot B:

72% > 70%

Therefore, Chatbot B does not satisfy the criterion.

Distance from 50%:

72 − 50 = 22 percentage points

- For System C:

58% < 70%

Therefore, System C satisfies the criterion.

Distance from 50%:

58 − 50 = 8 percentage points

Thus, System C is closest to the 50% guessing level among the three systems.

(c) Compute the improvement in fooling rate from ELIZA to System C in percentage points and as a ratio. Also compute the improvement from Chatbot B to System C in percentage points. **[5 Marks]**

ELIZA to System C
ELIZA fooling rate = 6%
System C fooling rate = 42%
Improvement:
42 − 6 = 36 percentage points
Ratio:
42/6 = 7
Therefore, System C's fooling rate is **7 times** that of ELIZA.
Chatbot B to System C
42 − 28 = 14 percentage points
Therefore, the improvement is **14 percentage points**.

(d) Interpret: the Turing Test measures "acting humanly". Explain why a high fooling rate does not prove that System C is thinking or acting rationally, using one idea from the foundations of AI (for

example the four approaches to defining AI, or the Chinese Room argument). **[4 Marks]**

- The Turing Test measures whether a machine can **act humanly** during a conversation.
- A high fooling rate does not necessarily prove that a system is genuinely thinking, understanding or acting rationally. The Chinese Room argument, for example, questions whether producing appropriate responses is sufficient to demonstrate genuine understanding.
- Therefore, System C's 42% fooling rate shows its performance in this particular conversational test. It does not by itself prove human-like thinking or rationality.

Question 4: Rational Route Choice under Uncertainty

Topic: Concept of rationality (expected performance, rational versus omniscient agent) | 15 minutes

A rational taxi agent must reach the airport. The deadline is 55 minutes. Its performance measure (utility) for a trip is:

$$U = 100 - T - 20 \cdot L$$

where T is the travel time in minutes and L = 1 if the taxi arrives after the 55-minute deadline, otherwise L = 0.

Route	Normal time (min)	Probability of disruption	Time if disrupted (min)
R1: Highway	30	0.30 (accident)	70
R2: City road	45	0.10 (signal failure)	60
R3: Ring road	50	0 (never disrupted)	50

The utility function is:

U = 100 − T − 20L

where:

- T = travel time
- L = 1 if the taxi is late
- L = 0 if the taxi is on time

The deadline is **55 minutes**.

(a) Compute the utility of each route in each outcome (normal and disrupted), clearly marking the late trips. **[4 Marks]**

R1 – Highway
Normal time = 30 minutes.
Since 30 ≤ 55:
L = 0
U = 100 − 30 = 70
During disruption, time = 70 minutes.
Since 70 > 55:
L = 1
U = 100 − 70 − 20 = 10
Therefore:
R1: Normal = 70, Disrupted = 10 (Late)
R2 – City Road

Normal time = 45 minutes:
U = 100 − 45 = 55
Disrupted time = 60 minutes:
U = 100 − 60 − 20 = 20
Therefore:
R2: Normal = 55, Disrupted = 20 (Late)
R3 – Ring Road
Time = 50 minutes and there is no disruption.
Since 50 ≤ 55:

U = 100 − 50 = 50
Therefore:
R3 Utility = 50 (On time)

(b) Compute the expected utility of R1, R2 and R3 and state which route the rational agent should choose. **[5 Marks]**

The formula is:
EU = P(Normal) × U(Normal) + P(Disruption) × U(Disruption)
R1
Normal probability:
1 − 0.30 = 0.70
Therefore:
EU(R1) = (0.70 × 70) + (0.30 × 10)
= 49 + 3 = 52
R2
Normal probability:
1 − 0.10 = 0.90

Therefore:
EU(R2) = (0.90 × 55) + (0.10 × 20)

= 49.5 + 2 = 51.5
R3
EU(R3) = 1 × 50 = 50
Therefore:

- **R1 = 52**
- **R2 = 51.5**
- **R3 = 50**

The rational agent chooses **R1 – Highway**, because it has the highest expected utility.

(c) A new percept arrives: a radio alert reports a major accident on the highway, raising the probability of disruption on R1 to 0.80. Recompute the expected utility of R1 and re-decide the route. **[4 Marks]**

The new alert increases the probability of disruption on R1 to **0.80**.

Normal probability:

1 − 0.80 = 0.20

Therefore:

EU(R1) = (0.20 × 70) + (0.80 × 10)

= 14 + 8

= 22

The other expected utilities remain:

R2 = 51.5

R3 = 50

Therefore, the new choice is:

R2 – City Road

The decision changes because the new percept changes the probability of disruption on R1.

(d) Suppose the agent chose the route in part (b) and the disruption did occur. Compute the utility actually obtained and state whether the agent was irrational. Distinguish between a rational agent and an omniscient agent. **[4 Marks]**

In part (b), the agent selected R1. If disruption occurs, the travel time becomes 70 minutes.

Since the taxi is late:

L = 1

Therefore:

U = 100 − 70 − 20 = 10

The actual utility is **10**.

The agent was **not irrational**. At the time of decision, R1 had the highest expected utility based on the available information.

A rational agent chooses the action that maximizes expected performance using available information. An omniscient agent would know the actual future outcome before making the decision.

Therefore, a rational agent can make a rational decision even if the final outcome is unfavorable.

(e) Interpret: explain why rationality depends on the percept sequence and the performance measure, and why part (c) changes the decision although the roads themselves did not change. **[3 Marks]**

- Rationality depends on the **percept sequence and performance measure**.
- Initially, R1 had an expected utility of 52, so it was selected. After receiving the accident alert, the probability of disruption increased to 0.80 and the expected utility of R1 decreased to 22.
- The roads themselves did not change, but the agent received new information. Therefore, its rational decision changed.
- This shows that an intelligent agent continuously uses new percepts and available information to select actions that maximize expected performance.

Rubrics for Calculation

Criteria	Weightage	Level 4 – Exemplary (100% of marks)	Level 3 – Proficient (75%)	Level 2 – Developing (50%)	Level 1 – Beginning (25%)
Problem Understanding	**20%**	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	**20%**	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	**25%**	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	**20%**	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	**15%**	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

Q. No. / Criteria	Problem Understanding (4)	Method Selection (4)	Solution Process (5)	Accuracy of Calculation (4)	Interpretation of Results (3)	Sub. Total (20)	Total %
1							
2							
3							
4							
Total							

Faculty In-charge	Course Coordinator	Program Director
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____